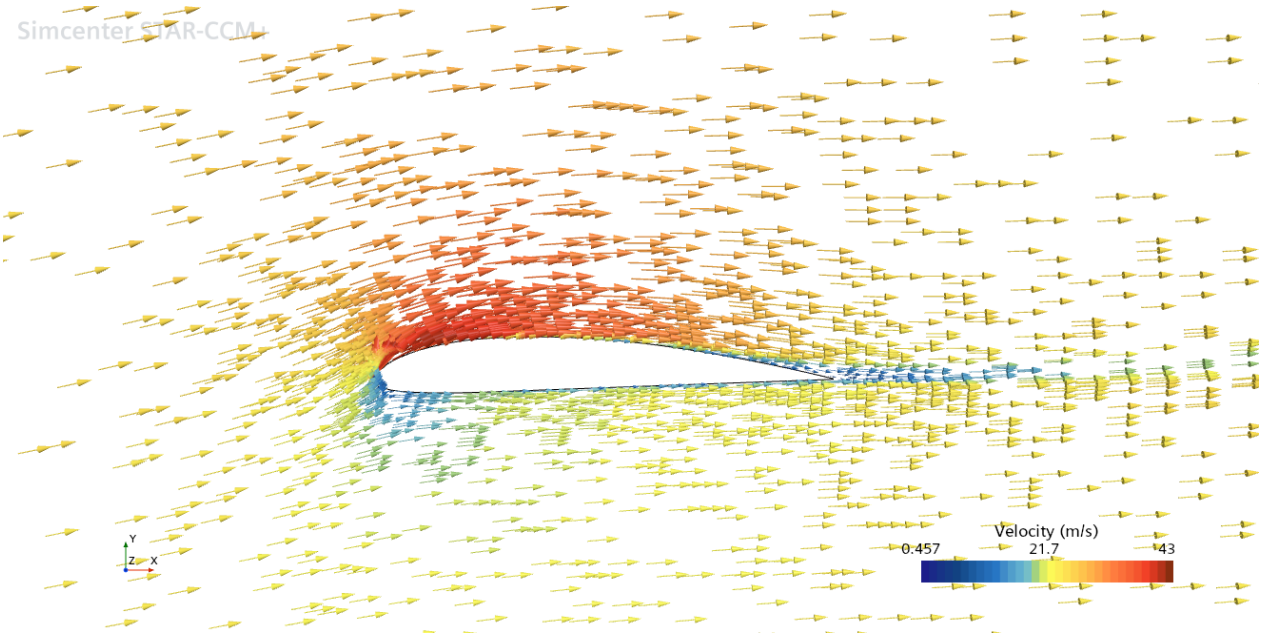
Glider Wing Study

Aerodynamic Analysis & Structural Dimensioning

Hamza Houir Alami | ENSAM Lille | 2025–2026

In collaboration with Étienne Roy

Supervisors: Prof. Antoine Dazin & Prof. Francesco Romano



CFD velocity field around Clark Y airfoil ($\alpha = 7°$, $Re \approx 2.2 \times 10^6$) — turbulent k–ε model, STAR-CCM+

1. Executive Summary

This report documents a three-part academic project on glider wing design conducted at ENSAM Lille during the 2025–2026 academic year. The project progresses from airfoil selection, through computational fluid dynamics (CFD) simulation, to structural dimensioning of the main spar.

Project	Glider Wing Study
Institution	ENSAM Lille (Arts et Métiers ParisTech)
Collaborator	Étienne Roy
Supervisors	Prof. Antoine Dazin (Part I) · Prof. Francesco Romano (Part II)
Period	November – December 2025
Tools	Python · STAR-CCM+ · Abaqus
Airfoil selected	Clark Y (finesse $f_{\max} \approx 108$–115)
Wing specs	$M = 599\,\text{kg}$, $V = 90\,\text{km/h}$, $L = 22\,\text{m}$, $c = 1.25\,\text{m}$
CFD model	k–ε turbulent, $Re \approx 2.2 \times 10^6$
Spar material	Carbon fibre, $e_{\min} = 0.20\,\text{mm}$ (hollow elliptical)

Three-part structure: Part I — Analytical airfoil selection (XFoil data, C_z analysis, finesse comparison). **Part II** — CFD study of the selected profile in STAR-CCM+: mesh sensitivity, physical model comparison (inviscid / laminar / turbulent). **Part III** — Structural dimensioning of the main spar: Euler–Bernoulli beam theory, material optimisation (Python), and FEM validation (Abaqus).

2. Project Context

The goal is to dimension the wing of a single-seat glider meeting the following specification sheet: total mass $M = 599\,\text{kg}$, cruise speed $V = 90\,\text{km/h}$ ($25\,\text{m/s}$), wingspan $L = 22\,\text{m}$, chord $c = 1.25\,\text{m}$.

The project was carried out in two parallel streams:

- **Aerodynamic stream** (Parts I & II): choose the best airfoil profile, compute the minimum incidence angle for level flight, and validate the choice via CFD simulation.
- **Structural stream** (Part III): pre-dimension the main spar that transmits aerodynamic loads to the fuselage, while minimising mass.

The Reynolds number at cruise conditions is:

$$Re = \frac{V \cdot c}{\nu} = \frac{25 \times 1.25}{1.42 \times 10^{-5}} \approx 2.2 \times 10^6$$

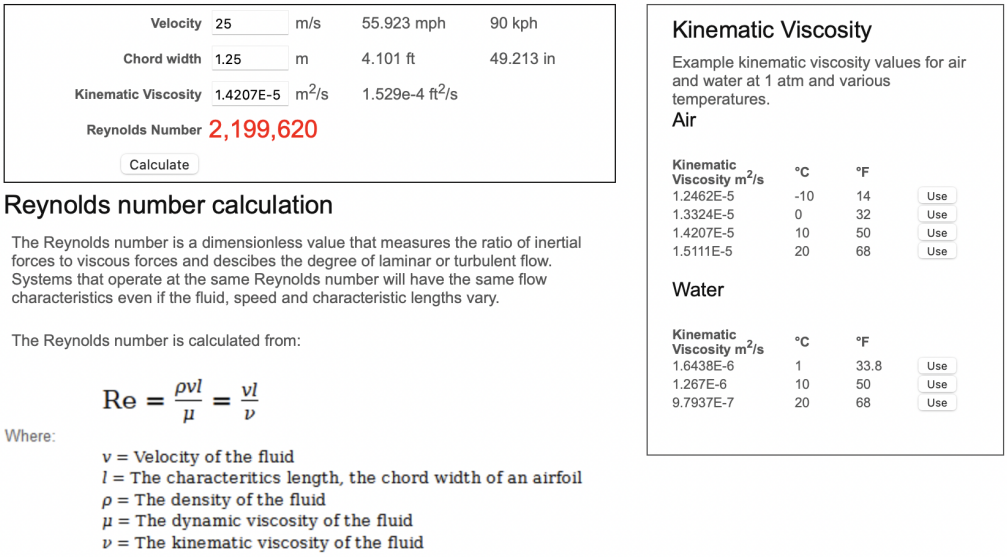


Figure 1. Reynolds number calculation at cruise conditions ($V = 25\,\mathrm{m/s}$, $c = 1.25\,\mathrm{m}$, $\nu = 1.42 \times 10^{-5}\,\mathrm{m^2/s}$).

At this Reynolds number the flow is turbulent, which governs both the profile selection and the CFD modelling strategy.

3. Part I — Airfoil Selection

3.1. Design constraint: minimum lift coefficient

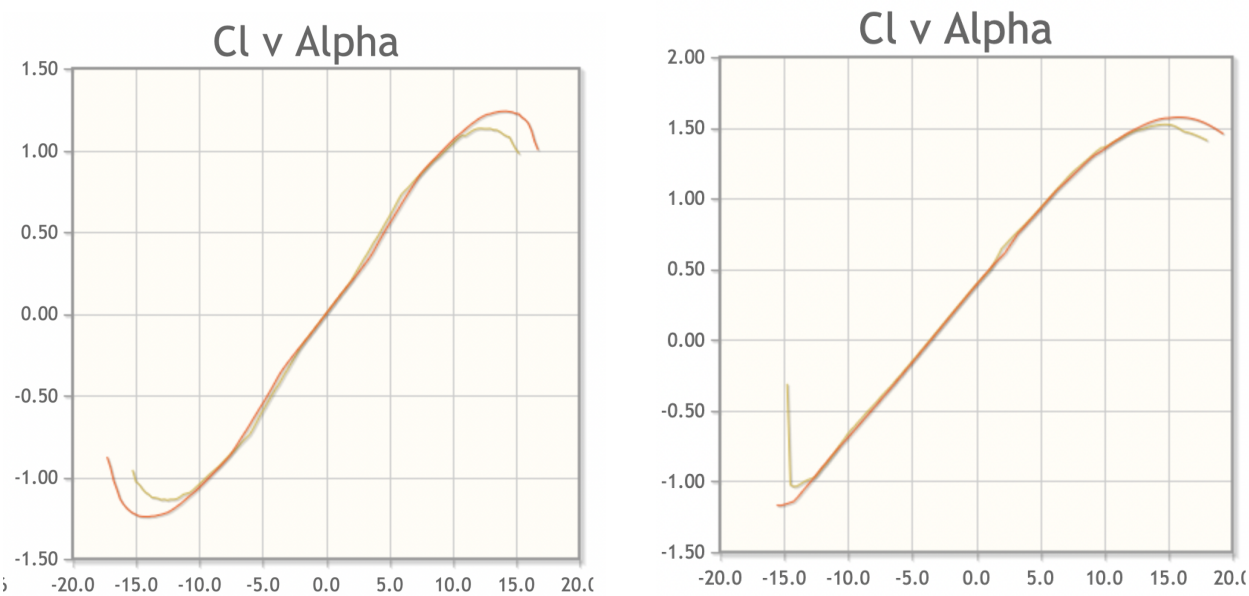
For level flight, lift must equal weight. Applying $L = \frac{1}{2}\rho V^2 S C_z$ with $L = Mg$, $S = L \cdot c$ and $\rho_{\text{air}} = 1.3\,\mathrm{kg/m^3}$:

$$C_{z,\min} = \frac{Mg}{\frac{1}{2}\rho V^2 S} = \frac{599 \times 9.81}{\frac{1}{2} \times 1.3 \times 25^2 \times (22 \times 1.25)} \approx \mathbf{0.41}$$

Any candidate airfoil must achieve $C_z \geq 0.41$ at the smallest possible angle of incidence, while maximising finesse ($f = C_L/C_D$).

3.2. RAF 27 vs. Clark Y — comparative analysis

Two profiles were compared using XFoil simulation data at $Re = 10^6$ (the curves converge for $Re \geq 10^6$, as verified in Section 2):



(a) C_z vs. α — RAF 27 airfoil **(b)** C_z vs. α — Clark Y airfoil

Figure 2. Lift coefficient curves for both candidate profiles (XFoil, $Re = 10^6$).

Parameter	RAF 27	Clark Y
$\alpha_{\min}$ for $C_z = 0.41$	4.00°	0.25°
Finesse at $\alpha_{\min}$	≈ 52	≈ 54–68
$f_{\max}$ (model 1 / model 2)	69.4 / 71.5	114.8 / 108.5
α at $f_{\max}$	6.25° / 7.5°	6.75° / 6.0°
Stall range	−12° to +13°	−14° to +14°

Selected profile: Clark Y. It achieves the required $C_{z,\min} = 0.41$ at only $\alpha = 0.25°$ (vs. 4.0° for RAF 27), offers a maximum finesse more than **50% higher** ($f_{\max} \approx 111$ vs. 70), and provides a wider stall-free operating range. All subsequent analyses use Clark Y.

4. Part II — CFD Simulation (STAR-CCM+)

4.1. Numerical setup

The Clark Y profile was simulated in STAR-CCM+ using the finite-volume method. The computational domain extends 10 chord lengths upstream and laterally, and 20 chord lengths downstream (C-Grid topology). The profile is a 2D extrusion. Boundary conditions: velocity inlet $V = 25\,\mathrm{m/s}$; pressure outlet P_{atm}; no-slip wall on the profile surface.

Three physical models were compared:

- **Inviscid (Euler):** no viscosity, no boundary layer.
- **Laminar:** molecular viscosity only (physically inappropriate at $Re \approx 2.2 \times 10^6$).
- **Turbulent k–ε:** RANS closure with turbulent viscosity.

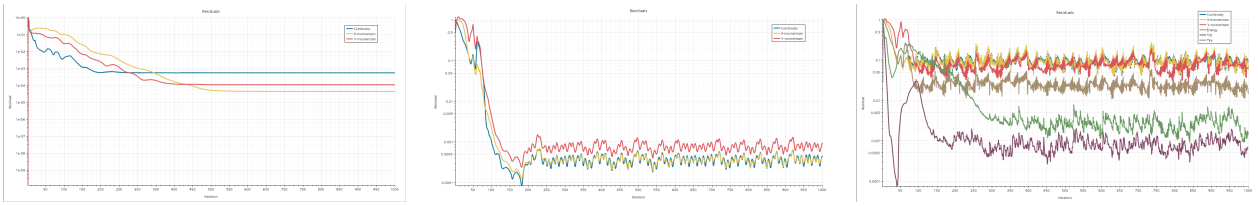
4.2. Mesh sensitivity study

Three mesh densities were tested (50, 100, 150 subdivisions along the chord) under turbulent conditions at $\alpha = 7°$:

Mesh	**Euler** C_L	**Laminar** C_L	**Turbulent** C_L
50 (coarse)	1.25	−0.025	1.09
100 (medium)	1.25	−0.075	1.09
150 (fine)	1.25	−0.075	1.09

The relative error between the medium and fine meshes is below 1% for C_L. **The 100-subdivision mesh is retained** as the optimal cost/accuracy compromise.

4.3. Physical model comparison and convergence



(a) Inviscid (Euler)
Stable plateau

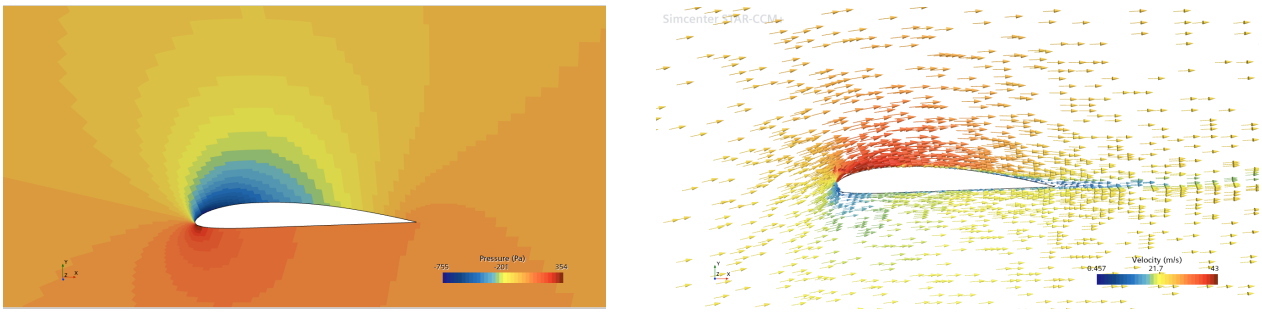
(b) Laminar
Periodic oscillations

(c) Turbulent k–ε
Converged after tuning

Figure 3. Residual convergence history for the three physical models.

The laminar model exhibits persistent sinusoidal oscillations — a sign of genuine physical unsteadiness (von Kármán shedding) that a steady solver cannot capture. The turbulent model required CFL reduction before converging; once stabilised, it produced physically consistent results.

4.4. Flow field visualisation



(a) Pressure field — k–ε ($\alpha = 7°$)

(b) Velocity field — k–ε ($\alpha = 7°$)

Figure 4. Turbulent simulation results: attached boundary layer, coherent suction zone, thin wake. $C_L \approx 1.075$, $Re \approx 2.2 \times 10^6$.

Model	C_L	C_D	**Physical validity**
Inviscid (Euler)	1.25	≈ 0	No
Laminar	< 0	> 0	Partial
Turbulent k–ε	1.075	> 0	Yes

The turbulent model correctly predicts $C_L \approx 1.075$ at $\alpha = 7°$ (theoretical reference: $C_L \approx 0.8$; the 35% overestimate is attributed to 2D assumptions and absence of explicit laminar-turbulent transition). At $\alpha = 15°$ the simulation fails to converge — consistent with stall onset (inherently unsteady).

5. Part III — Structural Dimensioning of the Main Spar

5.1. Euler–Bernoulli beam model

The main spar is modelled as a cantilever beam of length $L = 10\,\text{m}$ (half-span), loaded by distributed aerodynamic lift p_L and self-weight q_w (antagonistic loads):

$$q(e) = p_L - q_w = \frac{M_{\text{tot}} \cdot g}{2L} - \rho_{\text{mat}} \cdot g \cdot A(e)$$

Under the Euler–Bernoulli hypotheses (small deformations, plane sections remain plane), the maximum deflection and bending stress are:

$$v_{\max} = \frac{q(e)\,L^4}{8EI_{Gz}} \qquad \sigma_{\max} = \frac{M_{fz}^{\max} \cdot b}{I_{Gz}} \quad \text{(Navier formula)}$$

The spar cross-section is a **hollow ellipse** with semi-axes $a = 0.60\,\text{m}$ and $b = 0.15\,\text{m}$ (wall thickness e is the optimisation variable).

5.2. Material selection via Python parametric sweep

Five candidate materials were evaluated by sweeping e logarithmically from 0.1 mm to 50 mm against three simultaneous constraints: $v_{\max} \leq 0.30\,\text{m}$, $\sigma_{\max} \leq \sigma_y$, and $m_{\text{spar}} \leq 150\,\text{kg}$.

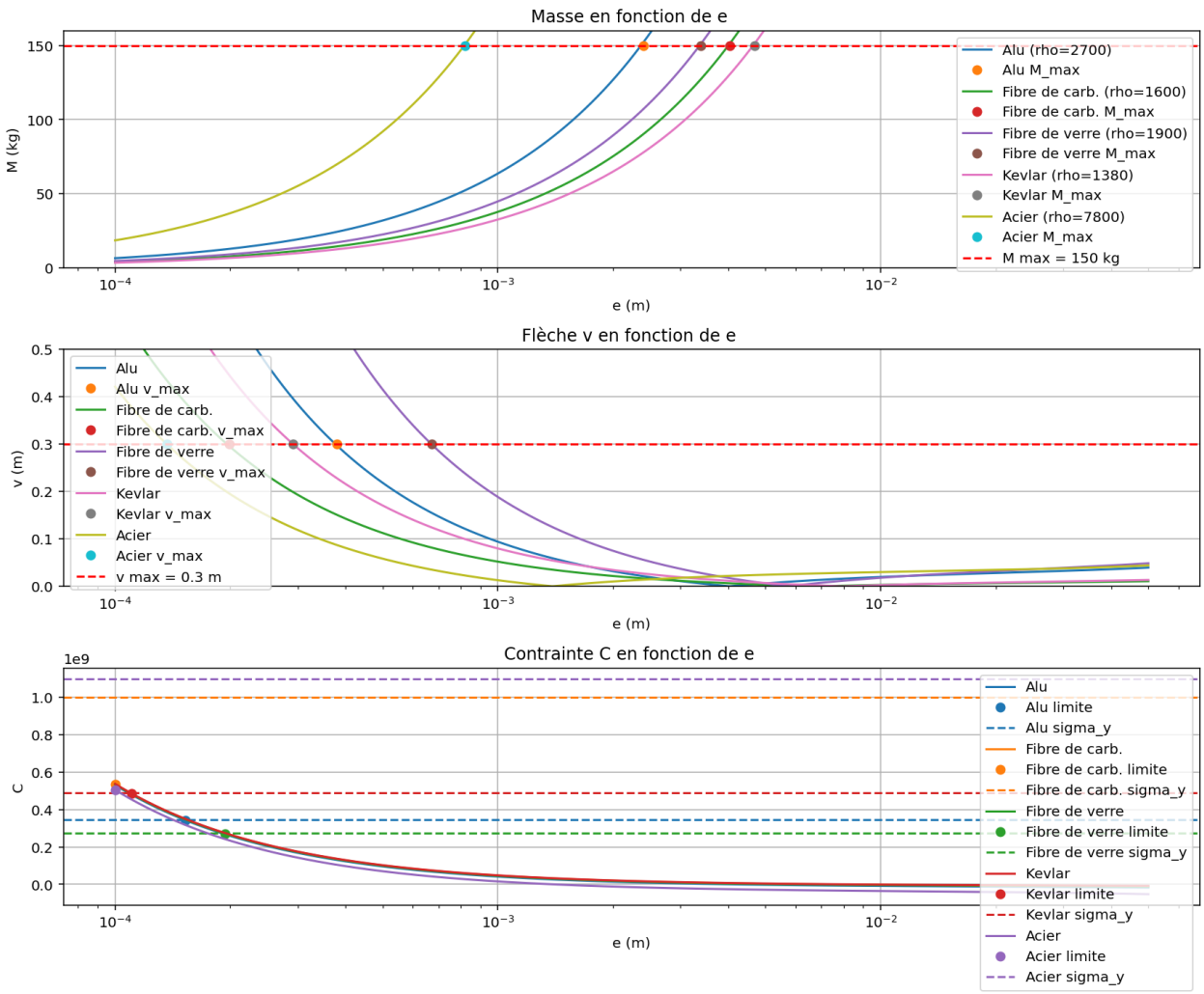


Figure 5. Python parametric sweep: mass (top), deflection (middle), and stress (bottom) vs. wall thickness e for all candidate materials. Dashed red lines mark the design constraints. Carbon fibre is the only material satisfying all three simultaneously with margin.

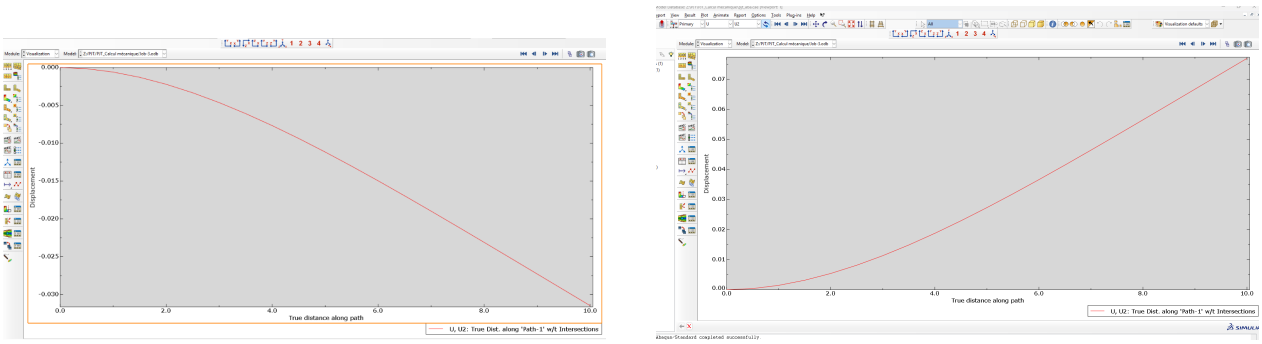
Material	$e_{\min}$ (deflection)	$e_{\max}$ (mass)	Verdict
Steel (aero)	Prohibitive	Very low	Too heavy
Aluminium 7075	Moderate	Moderate	Borderline
Glass fibre	Moderate	Moderate	Borderline
Kevlar	Moderate	Moderate	Borderline
Carbon fibre	$\approx 0.20\,\mathrm{mm}$	$\leq 4.04\,\mathrm{mm}$	Optimal

Selected material: carbon fibre. The deflection criterion is most restrictive, imposing $e \geq 0.198\,\mathrm{mm}$. The mass constraint allows up to $e \leq 4.04\,\mathrm{mm}$. Carbon fibre's exceptional specific stiffness (E/ρ) satisfies both constraints simultaneously, with a large margin — consistent with its near-universal use in real glider construction.

5.3. Abaqus FEM validation — hollow elliptical section

The analytical solution was validated in Abaqus using a beam finite-element model ($I_{Gz} = 7\times10^{-6}\,\mathrm{m}^4$, $e = 0.20\,\mathrm{mm}$, carbon fibre). Two load cases:

Load case	$M_{fz}^{\max}$	$v_{\max}$	**Comment**
On ground	3.57 kN·m	0.030 m	Self-weight only
In flight	9.00 kN·m	0.077 m	Lift − weight



(a) On ground — $v_{\max} = 0.030\,\mathrm{m}$ **(b)** In flight — $v_{\max} = 0.077\,\mathrm{m}$

Figure 6. Abaqus displacement along the spar (x-axis inverted: $x = 10$ = root, $x = 0$ = tip). Carbon fibre hollow ellipse, $e = 0.20\,\mathrm{mm}$.

Comparison with the analytical predictions revealed significant discrepancies: deflection error of 70% (ground) and 277% (flight), stress error up to 89% (ground). These errors point to **section ovalisation** at $e = 0.2\,\mathrm{mm}$: the Euler–Bernoulli model assumes the cross-section remains rigid, which fails for such a thin wall. The beam sections distort under load in Abaqus, leading to a completely different stress/deflection state.

5.4. Design revision — circular hollow section

To resolve the ovalisation issue, the design was revised to a **circular hollow section**. A Python optimisation identified the feasible region in the (R, e) plane:

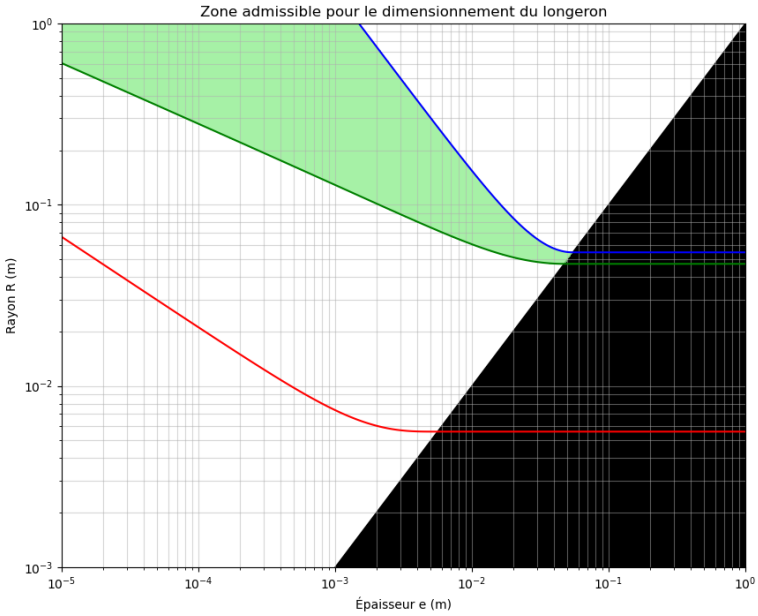
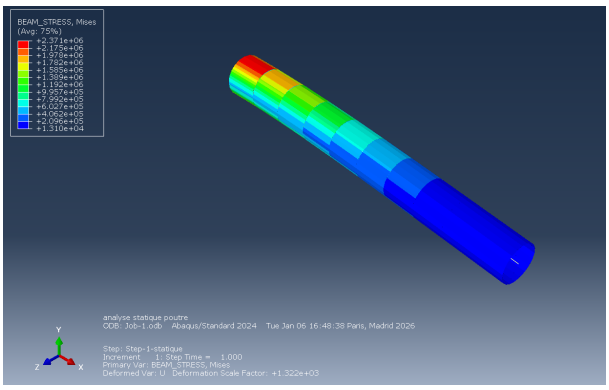
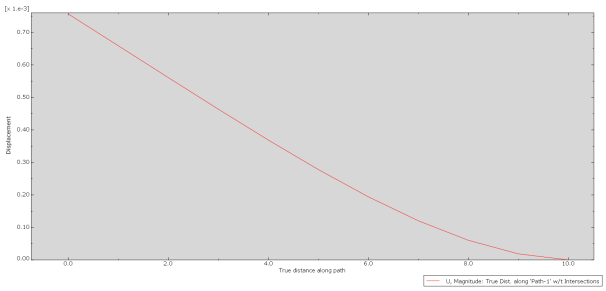


Figure 7. Feasible region (green) for the circular spar in the (R, e) plane. The retained design point: $R = 0.40\,\mathrm{m}$, $e = 1\,\mathrm{mm}$.



(a) Von Mises stress field $\sigma_{VM}^{\max} \approx 2.37\,\mathrm{MPa}$



(b) Displacement along spar $v_{\max} \approx 0.75\,\mathrm{mm}$

Figure 8. Abaqus results for the revised circular hollow section ($R = 0.40\,\mathrm{m}$, $e = 1\,\mathrm{mm}$, carbon fibre). The structure is very conservative: stress is well below the carbon fibre rupture limit ($\sigma_{\mathrm{rupt}} \approx 1{,}000\,\mathrm{MPa}$) and deflection is negligible.

6. Academic Connections

This project develops skills directly relevant to graduate-level engineering and operations research programmes.

6.1. Constrained optimisation in structural design

Part III is a **constrained optimisation problem** in disguise: find the minimum wall thickness e^* such that three constraint functions are satisfied simultaneously. In formal terms:

$$\min_e \; m(e) \quad \text{s.t.} \quad v_{\max}(e) \leq 0.30, \quad \sigma_{\max}(e) \leq \sigma_y, \quad m(e) \leq 150$$

The feasibility region in the $(e, \text{material})$ space is the intersection of three half-spaces. The optimal point lies on the boundary of the most restrictive constraint — here, the deflection constraint. This is the same structure as a linear programme with nonlinear objective functions.

6.2. Multi-criteria decision-making: airfoil selection

The airfoil selection in Part I is a **multi-attribute decision** over a discrete set (RAF 27, Clark Y) evaluated on multiple performance dimensions: minimum incidence, maximum finesse, stall range. Clark Y dominates RAF 27 on all criteria — a Pareto-optimal choice requiring no trade-off. In more complex design spaces, such decisions are formalised via TOPSIS, weighted scoring, or Pareto frontier analysis.

6.3. Numerical convergence and mesh sensitivity

The mesh sensitivity study in Part II is structurally a **convergence analysis**: as the discretisation parameter h (mesh element size) decreases, the numerical solution approaches the true PDE solution at a rate determined by the order of the scheme. Identifying the mesh at which results stabilise (asymptotic convergence region) is the numerical equivalent of detecting a fixed point in an iterative algorithm.

7. Key Results

Outcome	Value / Conclusion
Minimum lift coefficient	$C_{z,\min} = 0.41$ at cruise ($M = 599\,\text{kg}$, $V = 90\,\text{km/h}$)
Selected airfoil	Clark Y ($\alpha_{\min} = 0.25°$, $f_{\max} \approx 111$)
CFD model retained	Turbulent k–ε (100 subdivisions)
C_L at $\alpha = 7°$ (CFD)	≈ 1.075 (stable, mesh-independent)
Stall onset	$\alpha \approx 15°$ (simulation does not converge beyond)
Spar material	Carbon fibre (hollow elliptical section)
Minimum wall thickness	$e_{\min} \approx 0.20\,\text{mm}$ (deflection criterion)
Abaqus validation	In-flight: $M_{fz} = 9\,\text{kN·m}$, $v_{\max} = 77\,\text{mm}$

8. Limitations & Perspective

- **2D aerodynamic model:** Both the XFoil data and the STAR-CCM+ simulation are two-dimensional. Finite-span effects (tip vortices, induced drag) are not captured. A 3D simulation would require significantly greater computational resources.
- **Steady-state CFD:** The solver assumes a stationary flow; stall and post-stall dynamics are inherently unsteady (URANS or LES required).
- **Beam-only structural model:** The Euler–Bernoulli model neglects torsion, shear deformation, and cross-section distortion, all of which matter for thin-walled sections at $e = 0.2\,\text{mm}$.

- **No aeroelastic coupling:** Aerodynamic loads and structural deformation are treated independently. A complete design would require aeroelastic analysis (flutter, divergence).
- **Material isotropy assumption:** Carbon fibre is anisotropic; using isotropic properties is conservative but not representative of laminate design.

9. References

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